

DATA STRUCTURES

LAB PROGRAMS

20. Merge sort

```
#include <stdio.h>

void merge(int arr[], int l, int m, int r) {
    int i=l,j=m+1,k=0,temp[100];
    while(i<=m && j<=r) {
        if(arr[i] < arr[j])
            temp[k++] = arr[i++];
        else
            temp[k++] = arr[j++];
    }
    while(i<=m)
        temp[k++] = arr[i++];
    while(j<=r)
        temp[k++] = arr[j++];
    for(i=l,k=0;i<=r;i++,k++)
        arr[i] = temp[k];
}

void mergeSort(int arr[], int l, int r) {
    if(l<r) {
        int m=(l+r)/2;
        mergeSort(arr,l,m);
```

```
        mergeSort(arr,m+1,r);
        merge(arr,l,m,r);
    }
}

int main() {
    int arr[] = {38,27,43,3,9,82,10};
    int n = 7;
    mergeSort(arr,0,n-1);
    printf("Sorted Array:\n");
    for(int i=0;i<n;i++)
        printf("%d ",arr[i]);
    return 0;
}
```

Output

Sorted Array:

3 9 10 27 38 43 82

=== Code Execution Successful ===